User Manual for Tensile Expansion Microscopy (TExM) cell stretcher

Iris stretcher design and parts

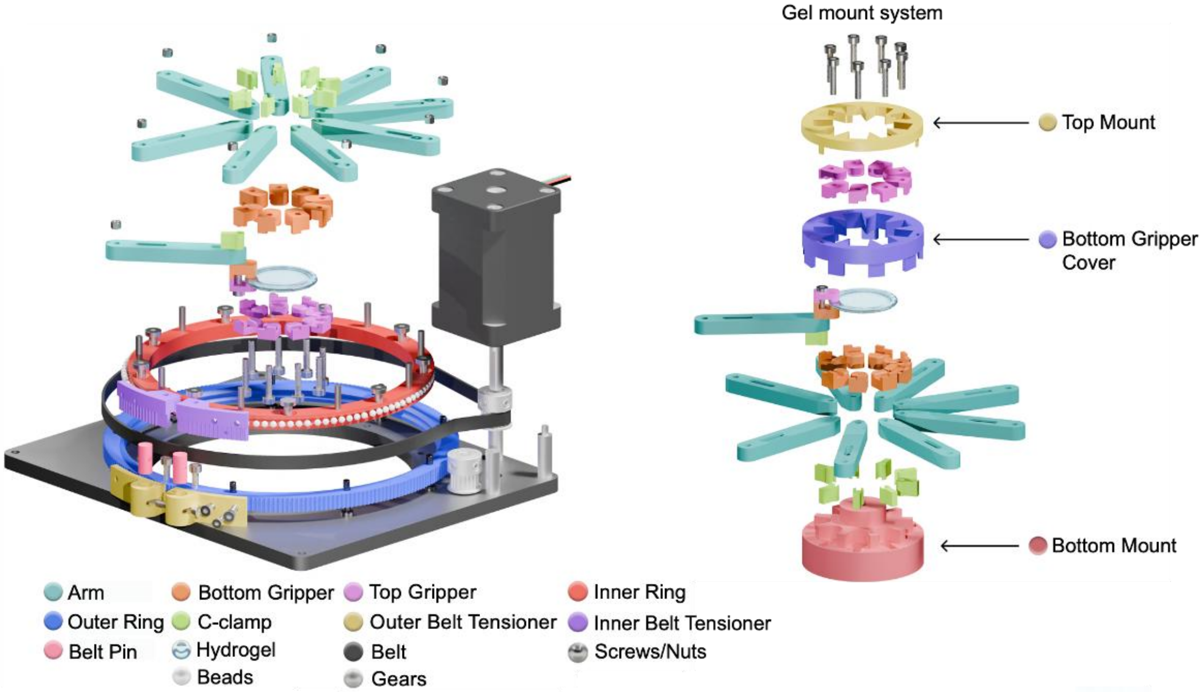


Table of Contents

Bill of Materials

Part Name	Quantity
Outer Ring	1
Inner Ring	1
Aluminum Base Plate	1
Arm	9
Bottom Gripper	9
Top Gripper	9
Inner Belt Tensioners	2
Outer Belt Tensioners	2
Belt	1
Heat Set Insert	9
Plastic Adhesion Glue	1
P60 Grid Sandpaper	1
Nylon Beads	2 packs
Timing Belt Pulley	2
Belt Pins	2
Aluminum Rod for Stepper Motor Stand	3
M3 Screw	30
M2 Screw	10
M5 Screw	1
M3 Threaded Rod	3

Machining instruction

The following parts need mechanical machining to be manufactured prior to assembly; Aluminum base plate and Aluminum Rod for Stepper Motor Stand.

1. Aluminum base plate

   Material: 6061 Aluminum sheet of MxN inch and X inch thick

   Machine: OMAX 5555 Waterjet Cutter, Power Drill, and T-handle tap wrench

   Procedure:

   - The base plate is designed using CAD software (refer to “Base plate.stl”).
   - The STL file needs to be exported as a DXF file prior to importing into a Waterjet machine.

- Preparing the Aluminum sheet by fixing onto a wooden board at its four corners then the board is clamped on to the machining bed.
- Operate the machine with care and assistance from trained personnel.
- Remove the cut Al sheet from the wooden board when finished.
- Due to machine tolerance, an additional drilling step is needed to ensure the cut holes have exact dimensions we need. A power drill with appropriate drill heads is used to ensure the hole size.
- To make threaded holes, T-handle tap wrench is used with appropriate drill bits (select the drill bits according to the screws pitch). Rotate the tap wrench to make threaded holes.

  Note: add some oil into the hole before tapping make it much easier

2. Aluminum Rod for Stepper Motor Stand

   Material: Hexagonal aluminum rod of X diameter and X inch long

   Machine: Horizontal Bandsaw, Drill Press, T-handle tap wrench

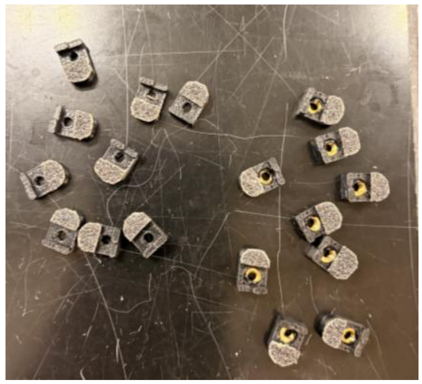
   Procedure:

- Measure the aluminum rod and mark the length you need using a marker
- Set up the horizontal bandsaw machine according to instruction or trained personnel
- Cut the aluminum rod at your markers using a horizontal bandsaw machine
- After cutting, make sure the cut surface is flat. If not, use Grinder to polish your part.
- After receiving small rods, use drill press machine with drill bit comparable to M3 screw hole size. Tap the rod with a tapping pen to mark the center location before drilling. Check the drill location prior to starting by lowering the drill head to align with the mark. If you see any small misalignment or bending of the drill head, realign the aluminum rod so it is exactly aligned as possible to prevent breakage of the drill head and unaligned drill hole. Once ready, make sure oil is added by brushing the surface of the rod and the drill bit, then starting the drill through the rod.
- After having Aluminum rods with holes in the center, you can now make threaded whole using a tap wrench like the previous section.

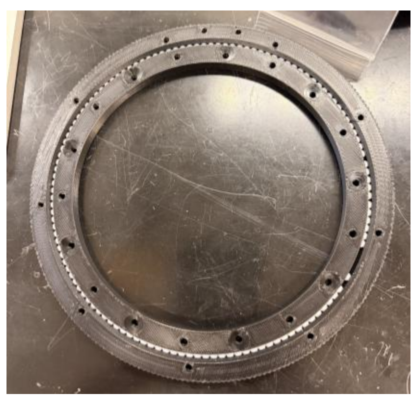
Caution: It is important to make sure that both surfaces of the rod are flat, and the threaded hole is straight and align. This will make the assembly much easier.

Assembly Procedure

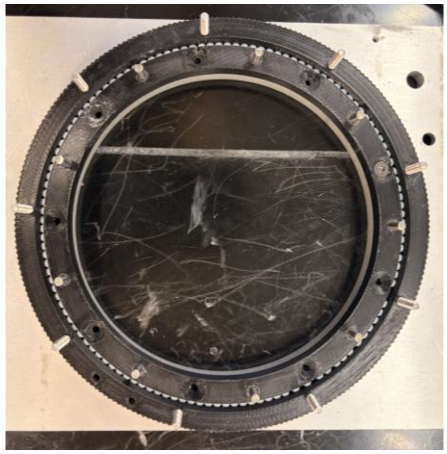
1. Make sure all components are ready according to the ‘Bill of material’.
2. Use a soldering iron to push the heat set to insert into its opening slot in the bottom grippers. Make sure the larger end of the heat set insert is leveled with the underside of the grippers.
3. Attach the sandpaper to the gripping area of both grippers using plastic adhesion glue. Then, trim the sandpaper to fit the shape of the grippers.



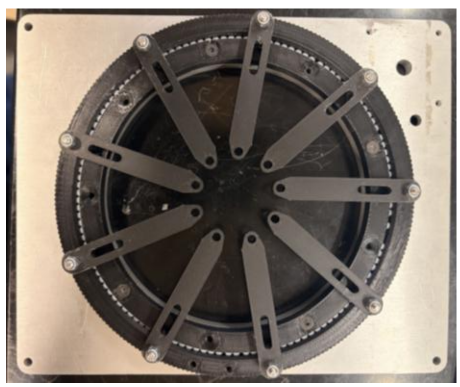
4. Press the nylon beads between the inner and outer rings by hand or use bottom of a screwdriver. Make sure the larger sides of the holes are facing up on the inner ring and face down on the outer ring.
   Caution: When pressing the nylon beads between the two rings, select one small section of the rings to do this rather than all around to avoid damaging surfaces of the rings which can cause obstructive and non-smooth rotation.



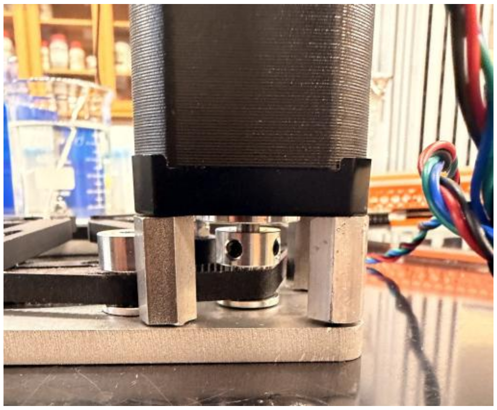
5. Drill ~1 mm into the pin holes, smaller holes in the inner ring, using a slightly larger bit, then hammer in the pins. Make sure the pins are hammered down until their height matches the arm’s height and are not bent or tilted.



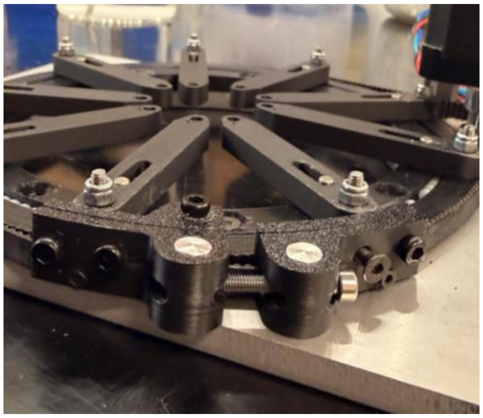
6. Screw into the outer ring from underneath using M3 screws. Attach the arms so that the screws and pins align with their slots/holes. Attach a nut to secure the arms onto the ring. Nut tightness readjustment is necessary after full assembly according to the hydrogel stretching test.
7. Before proceeding, check if the rotation of the ring is smooth without any skipping and overlapping of the nylon beads.



8. Drill the inner ring to the base. The smaller ends of the holes on the base should face up. Make sure to properly line up the holes before screwing.
9. Attach the pulley gear to the base. Insert a large screw through the underside of the base into the largest screw hole and place the pulley gear through the screw. Embed 2 small screws into the pulley gear to secure. It should spin securely. Another pulley gear attaches to the motor shaft and is secured with 2 small screws.
10. Attach the 3 motor stand rods to the motor using threaded rods. The threaded rods should be attached halfway into the motor screw holes and the aluminum motor stand rods. Then, secure the motor stand rod to the base plate using M3 screws.



11. Wrap the belt around the pulley gear and the outer ring.
12. Hammer the belt pins into the outer belt tensioners, aligning the screw holes. Cut the belt to the appropriate size. On one set, leave ~2 cm of the belt when aligned with the inner/outer belt tensioner. On the second set, leave space between the inner and outer belt tensioners to fit the remainder of the belt from the other side. Hole-punch the belt to match the size of the small screw hole on the outer belt tensioners.
13. Screw the inner/outer belt tensioner to the belt using M2 screws. Align the holes of the belt tensioner set to the outer ring. Connect both sides of belt tensioners together by inserting an M3 screw through the belt pins. Adjust the tightness to fit preferences. The image below shows the correct assembly of the belt tensioner.
    Note that there should be a space between the two sides of the tensioners to allow for future tightness adjustment since the belt can lose/stretch overtime.



14. The top and bottom frames of the control box are 3D printed and assembled together using M2 screws.
    Attach the control box to the iris by connecting the wires. Make sure the wires are connected in correct order or colors. A wire sleeve can be used to keep the wire together and prevent external damage.

General Cautions

- For machining parts, try to machine as precise as possible to avoid assembling issue such as misaligned holes, obstructed screw holes, and unlevel surfaces.
- Be sure to prepare extras of parts most susceptible to damage (grippers and inner/outer rings).
- To ensure immediate success, use caution to make sure you do not strip screws.
- When hammering in the pins, make sure they are not bent.
- When assembling the nylon beads, try to push the beads into place in one area to avoid damaging multiple areas of the inner and outer rings.
- Make sure to connect the wires in correct sequences/ colors.
- Do not force assemble if the parts are stuck, step back and investigate the issue to avoid damage to the parts.

Post-assembly test

Before using the stretcher on actual samples, make sure to perform a hydrogel stretch test experiment to ensure that the assembled iris stretcher runs smoothly and can stretch the hydrogel to full expansion factor. In addition, ensure the connectivity of the control panel of the control box by plugging in the power adapter and connect the ESP32 to your PC.

Sample preparation using hydrogel mounting system

1. Download and 3D print all STL parts related to the mounting system.
2. Prepare hydrogen sample: Cut the hydrogel in a circular shape of 30 mm (diameter) using a leather cutter.
3. Place all c-clamps into their slots in the bottom mount.
4. Insert bottom mount between the stretcher arms. The bottom mount should fit all 9 arms perfectly.
5. Place the bottom gripper cover over the arms and bottom mount from the top.
6. Lay each bottom gripper into its slot presents on the bottom gripper cover.
7. Lay the cut hydrogel on the center of the mounting system. Adjust the hydrogel so that its round edges are placed on the sandpaper area of all grippers.
8. Place the top mount on top of the hydrogel by aligning the three tiny legs of the top mount into their slots on the bottom gripper cover.
9. Once confirm that the hydrogel is locked in place with the top mount, proceed with screwing the top grippers into all bottom gripper.
   Caution: Recheck to make sure that all grippers secured/ gripped the hydrogel properly to avoid slipping or snapping of hydrogel while expanding. You can screw the grippers at its tightest.

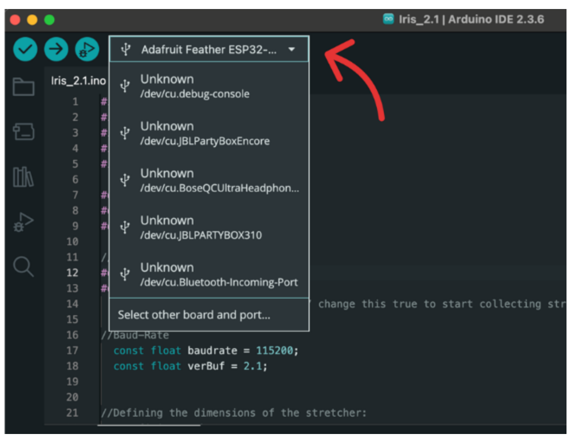
Stretcher assembly and Hydrogel mounting video can be found in Youtube.

Setting up the stretcher

Download all code files via our [Kisley lab Github page]

1. Connect the control box to your PC via USB-C cable. Open Arduino application on your PC – make sure you have serial monitor open up

- Follow the code uploading protocol written in Github page.
- If the code is already uploaded to the esp32 board, there is no need to upload the script again.
- Make sure you select the correct board and port by clicking the board selector menu in the top-left corner of the IDE (shown below). From the list, choose the entry that matches your connected device — for example, Adafruit Feather ESP32-S3 — and confirm that a valid port (e.g., /dev/cu.usbmodem… or COMx) is assigned.



2. Connect the PCB board (esp32) to the PC using USB-C and plug in the power adapter

3. Successful connection of the PCB to the PC should show “KISLEY/IrisStretcher” logo on the serial monitor with a set of command. The LCD on the PCB board should pop-up the menu.

- You can control the Iris from serial monitor following the commands shown (ex. Xgoto 4.2) OR control through the switches and turning knob following the menu shows on LCD screen on the PCB board.

- If desire, you can unplug the UCB-C cable from the PC. However, DO NOT unplug the power cord from the PCB, otherwise you need to redo step 2 again. This way, you can continue to use the Iris without PC connection.

Controlling the stretcher

Following are the menu and its function:

Menu	Function	Note
Xgoto	Move Iris arms to target expansion (Ex. 1.5 means 1.5x expansion)	Min value = 1.01 (If you use 1, the code will throw an error) Max value = 4.2 (fully stretch)
Xzero	Return Iris to original position	Useful when you are done with stretching and wants to bring the Iris back to original position to load a new sample
XsetZero	Set current position as position zero	It is recommended to perform XsetZero before you start the first stretch to let the board memorize starting position. Make sure the iris arms are unstretch.
Xcalibrate	Auto-calibrate the iris to find the zero position	The function is useful when the iris zero or max position is off. You can also unplug the power adapter and turn the iris by hand to the zero position (unstretch) and use XsetZero function.
Xspeed	Changes the speed of the expansion in approx cm/s	Default speed is set to 0.1 cm/sec. Speed less than 0.1 cm/sec can be set by typing into serial terminal in Arduino software interface. The control box will only show the slowest speed down to 0.1 cm/sec.